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LAB-TASK-6

People Detection Web App using Flask and OpenCV

Overview:
A real-time web application that detects people using the webcam, shows green boxes around detected people, and displays a live count on the web page. Detection uses HOG + SVM from OpenCV.

Requirements:

- Python
- Flask
- OpenCV (`opencv-python`)
- NumPy

File Structure:

- `app.py` → Main Flask app and detection logic
- `templates/index.html` → HTML template to display the video

How it Works:

- Flask serves a webpage with a video stream from the webcam.
- OpenCV detects people in each frame using HOG + SVM.
- Weak detections are filtered and overlapping boxes removed with Non-Maximum Suppression.
- People count is updated every few frames for stability.
- Annotated frames are streamed to the browser in real-time.

Usage:

1. Run `app.py`